

Pramod Chundury

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Mixed-methods HCI and accessibility researcher specializing in multimodal interaction, accessible data visualization, human-AI collaboration, and assistive technologies. My research investigates how sound, touch, speech, tactile interactions, and AI-enabled systems can support equitable access to complex information, particularly for blind and low-vision users. I have 7+ years of research experience across academic, industry, and cross-sector collaborations, including projects with Meta, Ericsson, Adobe Research, the Maryland State Department of Education, and the Library of Congress. My work combines interviews, co-design, usability studies, experiments, and system-building, with publications in ACM CHI, ACM ASSETS, and IEEE VIS.

Education

Ph.D. in Information Studies

Aug 2018 - May 2025

University of Maryland, College Park, MD, USA

Dissertation: Designing Multimodal Touchscreen Interactions For Accessible Data Visualization Supporting Blind Users

Advisors: Dr. Jonathan Lazar and Dr. Niklas Elmqvist

M.S. in Human-Computer Interaction

Aug 2015 - May 2017

University of Maryland, College Park, MD, USA

Mentors: Dr. Marshini Chetty and Dr. Leah Findlater

B.E. in Computer Science

Jun 2007 - Apr 2011

Anna University, Chennai, TN, India

Employment

Data Scientist/Visualization Researcher

Jan 2026 - Present

Data and Design, LLC, Davidsonville, MD, USA

Research Associate (UMD Language Science Center)

Apr 2026 - Jul 2026

University of Maryland, College Park, MD, USA

Mixed-Methods Researcher (College of Information)

Jun 2025 - Dec 2025

University of Maryland, College Park, MD, USA

Graduate Research Assistant

Sep 2020 - May 2025

University of Maryland, College Park, MD, USA

Graduate Teaching Assistant

Sep 2019 - Aug 2020

University of Maryland, College Park, MD, USA

Graduate Research Assistant

Aug 2018 - Aug 2019

University of Maryland, College Park, MD, USA

User Experience Researcher

Apr 2018 - Jul 2018

Meta Inc, Menlo Park, CA, USA

Research Assistant

Sep 2017 - Mar 2018

University of Washington, Seattle, WA, USA

User Experience Intern

May 2016 - Dec 2016

Ericsson, Manchester, NH, USA

Graduate Research Assistant

Mar 2016 - Dec 2016

University of Maryland, College Park, MD, USA

Application Developer

Jun 2011 - May 2015

Deloitte Consulting, Bengaluru, KA, IND

Research Experience

Fostering Inclusivity through Technology for Mixed-Neurotype Teams University of Maryland - Language Science Center	Apr 2026 - Jul 2026
Enhancing Accessibility of Visual Cultural Heritage University of Maryland and Library of Congress	Jun 2025 - Dec 2025
Accessible Data Visualization University of Maryland	Jul 2020 - May 2025
Designing Accessible Data Science and CS Courses University of Maryland, Maryland State Department of Education, Data A11y	Jan 2022 - Sep 2023
AI for Supporting Creative Collaboration University of Maryland and Adobe Research	May 2021 - Dec 2021
Designing and Evaluating Visual Analytics Tools for Medical Research Professionals University of Maryland	Sep 2020 - Dec 2021
Generating Textual Narratives for Causal Visualization University of Maryland and Virginia Tech	Jan 2020 - May 2020
Exploring Personal Tracking in Groups for Amateur Sports Teams University of Maryland	Aug 2018 - Dec 2019
Touchscreen Accessibility for People with Motor Impairments University of Maryland and University of Washington	Sep 2016 - Mar 2018
User Perceptions of Privacy and Security Surrounding Personal Drones University of Maryland	Mar 2016 - May 2017

Publications and Talks

Conference, Journal and Workshop Articles (Peer-reviewed)

2026	Exploring Tactile Chart Authoring for Refreshable Tactile Displays: Lessons from a Co-Design Case Study. Pramod Chundury, J. Bern Jordan, Yasmin Reyazuddin, Niklas Elmqvist, and Jonathan Lazar. <i>2026 IEEE Workshop on Accessible Data Visualization (AccessViz).</i>
2026	“Metadata is wonderful, but there’s a coldness to it”: exploring the visual accessibility of cultural heritage through participatory design. Victoria Van Hyning and Pramod Chundury. <i>Information Research an International Electronic Journal (iConference).</i>
2026	Sound, Touch, or the Full Monty? A Comparative Study of Data Accessibility Techniques for Blind Users. Pramod Chundury, J. Bern Jordan, Yasmin Reyazuddin, Niklas Elmqvist, and Jonathan Lazar. <i>ACM Transactions on Accessible Computing (TACCESS).</i>
2024	Understanding the Visualization and Analytics Needs of Blind and Low-Vision Professionals. Pramod Chundury, Urja Thakkar, Yasmin Reyazuddin, J. Bern Jordan, Niklas Elmqvist, and Jonathan Lazar. <i>SIGACCESS Conference on Computers and Accessibility (ACM ASSETS).</i>
2023	TactualPlot: Spatializing Data as Sound using Sensory Substitution for Touchscreen Accessibility. Pramod Chundury, Yasmin Reyazuddin, J. Bern Jordan, Jonathan Lazar, and Niklas Elmqvist. <i>IEEE Transactions on Visualization and Computer Graphics (IEEE VIS).</i>

- 2022 **Contextual In Situ Help for Visual Data Interfaces.**
 Pramod Chundury, Mehmet Adil Yalcin, Jonathan Crabtree, Anup Mahurkar, Lisa M Shulman, and Niklas Elmqvist.
Information Visualization.
- 2021 **Towards Understanding Sensory Substitution for Accessible Visualization: An Interview Study.**
 Pramod Chundury, Biswaksen Patnaik, Yasmin Reyazuddin, Christine Tang, Jonathan Lazar, and Niklas Elmqvist.
IEEE Transactions on Visualization and Computer Graphics (IEEE VIS).
- 2021 **Once Upon A Time In Visualization: Understanding the Use of Textual Narratives for Causality.**
 Arjun Choudhry, Mandar Sharma, Pramod Chundury, Thomas Kapler, Derek Gray, Naren Ramakrishnan, and Niklas Elmqvist.
IEEE Transactions on Visualization and Computer Graphics (IEEE VIS).
- 2019 **Motor Accessibility of Smartwatch Touch and Bezel Input.**
 Meethu Malu, Pramod Chundury, and Leah Findlater.
ACM SIGACCESS Conference on Computers and Accessibility (ACM ASSETS).
- 2018 **Exploring Accessible Smartwatch Interactions for People with Upper Body Motor Impairments.**
 Meethu Malu, Pramod Chundury, and Leah Findlater.
Conference on Human Factors in Computing Systems (ACM CHI).
- 2017 **Spiders in the Sky: User Perceptions of Drones, Privacy, and Security.**
 Victoria Chang, Pramod Chundury, and Marshini Chetty.
Conference on Human Factors in Computing Systems (ACM CHI).

Edited Publications

- 2025 **Building Multimodal Chart Accessibility for Blind and Low-Vision Individuals in the Prompt-Driven Era.**
 Pramod Chundury.
ACM XRDS (Crossroads).

Doctoral Colloquium

- 2023 **Designing Multimodal Touchscreen Interactions For Accessible Data Visualization Supporting Blind Users.**
 Pramod Chundury.
IEEE VIS Doctoral Colloquium.

Invited Talks

- 2026 **Sound, Touch, or the Full Monty? A Comparative Study of Data Accessibility Techniques for Blind Users.**
 ANZAGG Refreshable Tactile Display Special Interest Group.
- 2026 **Making Data Understandable: Designing Accessible and Multimodal Visualizations.**
 Kogod School of Business, American University.
- 2025 **TactualPlot: Spatializing Data as Sound using Sensory Substitution for Touchscreen Accessibility.**
 3rd Korea Visualization Workshop (K-VIS 2025).
- 2024 **TactualPlot: Spatializing Data as Sound using Sensory Substitution for Touchscreen Accessibility.**
 Human-Computer Interaction Lab Symposium, University of Maryland.
- 2019 **Personal Tracking in Groups: Exploring Tracking Behavior in Sports Teams.**
 Annual Physical Cultural Studies Graduate Student Conference, University of Maryland.

Press Coverage

- 2026 **Monarch is Making a Difference: Comparative Study Explores Statistics and Scatter Plots.**
American Printing House
- 2022 **Expanding the Possibilities for Accessible Data Visualization.** *UMD iSchool*
- 2020 **PhD Student Spotlight: Pramod Chundury.** *UMD iSchool*

Teaching Experience

Teaching Assistant, University of Maryland, College Park

INST 327 – Database Design and Modeling	Summer 2020
INST 127 – Introduction to Programming (Python)	Spring 2020
INST 462 – Introduction to Data Visualization	Fall 2019

Guest Lecturer, University of Maryland, College Park

INST 760 – Data Visualization (Accessible Data Visualization)	Fall 2023
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Awards and Recognition

Travel Grant for Doctoral Colloquium (IEEE VIS)	2023
Research Improvement Grant (University of Maryland)	2019
HCIM Travel Award (University of Maryland)	2017
Beyond These Walls Travel Award (University of Maryland)	2017
Outstanding Award for Exceptional Work (Deloitte Consulting)	2014
Applause Award for Innovation (Deloitte Consulting)	2014
Applause Award for Client Service (Deloitte Consulting)	2014

Special Recognition for Outstanding Reviews

ACM CHI (3 awards)	2026, 2024
ACM DIS (2 awards)	2026, 2025

Technical Skills

Languages: Python, JavaScript, HTML/CSS, SQL, C/C++, Swift, ABAP.

Frameworks & Libraries: D3.js, Vega-Lite, Matplotlib, Flask, Apache Cordova, Web Audio API, Pandas, OpenCV, scikit-learn, PyTorch.

Research and Product: Usability testing, co-design, interviews, experiments, survey design, heuristic evaluation, qualitative coding, thematic analysis, research synthesis, and prototyping.

Tools: Git, Figma, Tableau, UserTesting, UserZoom, SPSS, LaTeX, VS Code, Android Studio, Xcode, LLM CLIs.

Accessibility and Multimodality: WCAG, screen-reader-aware design, sonification, tactile interfaces, refreshable braille displays, accessible data visualization.

Academic Service

Organizing/Program Committees

Program Committee Member - ACM ASSETS	2026
Paper Chair - 3rd Workshop on Accessible Data Visualization (IEEE VIS)	2026

Invited Member - ANZAGG Refreshable Tactile Display Special Interest Group	2025 - Present
Paper Chair - 2nd Workshop on Accessible Data Visualization (IEEE VIS)	2025
Co-Chair - 1st Workshop on Accessible Data Visualization (IEEE VIS)	2024

Peer Reviewer

ACM CHI; ACM TOCHI; ACM DIS; ACM TACCESS; ACM ASSETS; Journal of Visual Impairment & Blindness; Universal Access in Information Society; IEEE VIS, IEEE TVCG; EuroVis; Journal of Computer Languages; Scientific Reports	2019 - Present
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Student Volunteer

Human-Computer Interaction Lab Symposium, University of Maryland, College Park	2016 - 2024
ACM Designing Interactive Systems (DIS)	2019

Mentoring

Enric Hongxiao Jiao, CS undergraduate, University of Maryland	2024
Urja Thakkar, MS in HCI, University of Maryland	2024
Christine W Tang, Poolesville High School	2019
Tyler Pavay, CS undergraduate, University of Maryland	2019
Tyson Nguyen, Information Science undergraduate, University of Maryland	2019
ABAP programming mentor, Deloitte Consulting, India	2012 - 2015
Math and English tutor, Zilla Parishad High School, India	2012